Andy Yang

ay.heong@outlook.com | 805-637-9244 | https://www.linkedin.com/in/andy-h-yang | https://github.com/ayheong

Professional Summary

Graduate computer science student with dual bachelor's degrees in Computer Science and Statistics, with experience spanning full-stack development, machine learning, and applied AI. Comfortable working across the stack, from training and deploying predictive models to building cross-platform applications and backend systems using Python, TypeScript, and modern frameworks.

Education

University of Texas at Austin
Master of Science in Computer Science. GPA: 3.8. — *Expected May 2027*

University of California, Davis
Bachelor of Science in Computer Science. GPA: 3.6. — *2021–2025*

University of California, Davis
Bachelor of Science in Statistics. — *2021–2025*

Relevant Coursework: Machine Learning, Deep Learning, Software Engineering, Database Systems, Big Data & Statistical Computing, Data Visualization

Projects

Neatnet — https://github.com/ayheong/neatnet
TypeScript | Tauri | React

- Built a cross-platform desktop app that uses AI to generate, preview, and safely apply bulk file reorganization plans with an integrated user review process.
- Engineered a topological move scheduler that detects dependency cycles and resolves them via temporary staging paths, preventing data loss on batched filesystem operations.
- Designed a dual AI backend supporting both Claude API and local Ollama models, enabling fully offline operation with no external dependency.
- Implemented structured JSON prompting with response validation and path normalization to safely translate LLM output into executable file operations.

Personal Finance App — https://github.com/ayheong/personal-finance-app
Python | BERT | Flask | MongoDB

- Developed a full-stack web application that transforms bank and credit card statement data into categorized expenses and budgeting insights.
- Created and labeled a dataset of 2,000+ financial transactions, fine-tuned and evaluated a BERT classifier, and integrated regex-based rules to categorize spending across 11 distinct expense categories.
- Integrated the model into a Flask and MongoDB backend with JWT-secured APIs, enabling real-time transaction classification and interactive budgeting dashboards.

ATP Tennis Match Predictor — https://github.com/ayheong/tennis-pred
Python | LightGBM | FastAPI | Docker

- Engineered an end-to-end machine learning pipeline to predict ATP tennis match outcomes across a dataset of 76,000+ historical matches.
- Built domain-specific features including head-to-head performance, Bradley-Terry skill ratings, and surface-specific win rates.
- Trained, evaluated, and optimized an ensemble of LightGBM models using Optuna for hyperparameter tuning; deployed the service as a FastAPI application containerized with Docker.

Car Racing with Reinforcement Learning — https://github.com/ayheong/DQN_PPO_gym_2d_car_racing
Python | PyTorch

- Developed custom reinforcement learning agents in PyTorch applied to Gymnasium's 2D Car Racing environment, comparing on-policy vs. off-policy approaches to evaluate model trade-offs.
- Tuned hyperparameters and applied training techniques including experience replay, target networks, and policy clipping; assessed training efficacy through reward/loss curves and environment playback.

Skills

Languages: Python, SQL, R, TypeScript, Java, C++
Frameworks: PyTorch, TensorFlow, Scikit-learn, Flask, FastAPI, React, Tauri
Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Jupyter
Developer Tools: Git, GitHub, Docker, Linux, MongoDB